Measurement Uncertainty
CS18 SPL

Abstract

of the detailed Measurement uncertainty budget
(Part 6 SPL)
for measurand sound pressure with sinusoidal excitation

(Calibration by comparison in a pressure chamber)

Author: Bühn / Begoff
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Basic theory

The following document is based on

[M1] SPEKTRA: Measurement uncertainty budget for the measurement of sound pressure with sinusoidal excitation (calibration by comparison in a pressure chamber), Part 6 System CS18 SPL, edition 2010

and the bibliographical references made in this document

1. Calibration methods, calibration job and calibration item

Which calibration method to apply depends on the kind of calibration item. If the calibration item is a **microphone**, calibration is performed using the **method of comparison**, either by substitution (sequential comparison) or by direct comparison by means of an acoustical coupler (parallel comparison). In either case the calibration item is compared with a reference microphone (reference standard).

When performing **calibration by substitution**, the item under test and the reference standard are exposed to the same sinusoidal sound pressure **one after another**. This sound signal is generated mechanically by means of a pistonphone. The ratio of microphone output voltages measured on a channel of the calibration system corresponds to the ratio of the transfer coefficients of item under test and traceable reference standard.

If the calibration item is a **microphone**, the calibration job is to determine either the transfer coefficient S_X of microphone and associated preamplifier or the so-called open-circuit transfer coefficient S_{0X} of the microphone cartridge alone. In the latter case, impedance matching and preamplification are carried out by means of a special microphone amplifier that is a component part of each CS18 SPL Standard measurement systems.

If the calibration item is a **Sound level meter**, calibration is performed again by the above mentioned **method of comparison**. In this case the characteristics of the entire measuring chain are determined including the indicator.

If the calibration item is an acoustical **Calibrator**, first the produced sound pressure level and some other quantities (frequency, total harmonic distortion) are determined, using the reference microphone of the standard measurement system CS18 SPL. In an optional step the calibration item can be substituted for a calibrator used as a reference standard (e. g., an acoustical calibrator) and the sound pressures produced by the calibration item and the reference standard are compared with one another.

The **1st calibration item** is a ½" measurement microphone type MTG MK 221, the **2nd calibration item** is a ¼" measurement microphone type MTG MK 301 with preamplifier MTG MV203. In either case the **calibration job** is to determine the transfer coefficient S_X of the microphone as a function of frequency.

The **3rd calibration item** is a sound level meter with detached microphone. Here the **calibration job** is to generate a well-defined sound pressure and to calibrate the entire measurement chain including its indicator. Calibration is carried out at fixed frequencies 250 Hz and 1000 Hz using the two sound calibrators (reference standards 3 and 4).

For a detailed performance specification and description of the type SPEKTRA CS18 SPL **Standard measurement system** and its component parts used for performing the above mentioned calibration jobs, please refer to the current Technical data sheet [19].

As far as the measurement uncertainty budget is concerned, the analog components are the crucial points. There are two measuring channels in each measurement system, each of which is equipped with two input converters, selectable by multiplexer, to enable condenser microphones (Lemosa connectors) or other sources of ac voltage (BNC connectors) to be connected directly to the system.

The characteristics of each input, including the following programmable voltage amplifier, are tested for the first time in the final test run after production and later at regular intervals by means of traceable precision measuring instrumentation. The results are recorded and stored on an allocated memory. Later, when using the system, the calibration software CS18Win will have access to this memory and correct all bias errors of the analog hardware that have been saved before.

Electrical tests

In addition to the mentioned acoustical tests, sound level meters are required to pass special electrical tests according to DIN EN 61672-3. These tests and the uncertainties involved are described in detail in a separate MUB.

2. Model equation

2.1 Sound

As required, minimum measurement uncertainty is verified in theory by means of the following detailed measurement uncertainty budget and in practice by measuring test samples with precisely known transfer characteristics.

A comprehensive description of the mathematical procedures can be found in [M1], so the details of derivation need not be repeated in this place. In the end, the detailed expression – defining equation with *best* estimates – is given by the following equation:

$$S_X = S_S * G * R * K_{FPC} * K_{IS} * K_{IG} * K_{IR} * K_{TD} * K_{PDM} * K_{PDC} * K_{HUM} * K_{FFC} * K_{POL} * K_F * K_{SUB} * K_{AD} * K_{RES} \quad (1)$$

If measurements are taken in succession with the same setting of gain G, the defining quantities G and R are equal to 1,0 and there are no uncertainty contributions $w(G)$ and $w(R)$, as the instability with time and the uncertainty of the substitution are already covered by the correction factors K_{IG} , K_{IR} und K_{SUB}.

When calibrating sound calibrators by the substition method or by absolute measurement, substitute the defining quantities S_X and S_S for the sound pressures p_X and p_S. For the defining quantities G and R, the above notice applies.

The names of members are listed in the following survey.

Y **= *S*x** **= *S*0** oder **= *p*x**	Result: transfer coefficient of calibration item (KG) - microphone with preamplifier - microphone without preamplifier or result: sound pressure of calibration item (KG) or generated sound pressure	1	
$X_1 = S_S$ or $= p_S$	- transfer coefficient of reference standard (reference microphone) or - sound pressure of reference standard (pistonphone)	2	
$X_2 = G$	Ratio of amplifier gain coefficients $G = G_S / G_X$		
$X_3 = R$	Ratio of output voltages $R = V_X / V_S$		
$X_4 = K_{FPC}$	Correction factor Free field-pressure conversion		3
$X_5 = K_{IS}$	Correction factor Instability of reference standard with time		
$X_6 = K_{IG}$	Correction factor Instability of amplifier gain coefficient with time		
$X_7 = K_{IR}$	Correction factor Instability of AD converter with time		
$X_8 = K_{TD}$	Correction factor Temperature deviation		
$X_9 = K_{PDM}$	Correction factor Pressure deviation		
$X_{10} = K_{PDC}$	Correction factor Pressure measurement		
$X_{11} = K_{HUM}$	Correction factor Humidity deviation		
$X_{12} = K_{FFC}$	Correction factor Deviation from true free-field conditions		
$X_{13} = K_{POL}$	Correction factor Polarization voltage		
$X_{14} = K_F$	Correction factor Frequency		
$X_{15} = K_{SUB}$	Correction factor Substitution		
$X_{16} = K_{AD}$	Correction factor Application of reference standard (for calibrators only)		
$X_{17} = K_{RES}$	Correction factor Residual effects		

Table 1: Names of the product terms. The different colours identify
- ☐ defining quantities
- ☐ fixed influence variables (determined by the system)
- ☐ influence variables whose values may have to be determined by the user
- ☐ influence variable is relevant only if reference standard was calibrated in a free field

[1] Result quantity
[2] *N'* Input quantity for determining the result quatity (defining quantities)
[3] *N* Input quantity for determining measurement uncertainty (influence variables)

The partial derivations of Equ. (5) with respect to the each input variable x_i are the uncertainty contributions $w_{S_S}(S_X), w_R(S_X), w_G(S_X)$ etc. (refer also to the general budget table 4.1) which, neglecting covariance, can be summed up according to

$$w(S_X) = \sqrt{\sum_{i=1}^{N} w_i^2}$$

to obtain Total relative measurement uncertainty. Introducing the Coverage factor (k=2), this quantity can be further processed to yield the Expanded measurement uncertainty *W(S_X)* for the magnitude of S_X. The individual contributions will be dealt with subsequently in more detail.

2.2 Frequency

In a detailed representation – defining equation with *best* estimates – equations (1) to (4) can be summed up, yielding the following expression for the measurement of the frequency of calibrators:

$$f_X = f_S * K_{IK} \tag{6}$$

where f_X is the result, f_S is the frequency determined by means of the frequency meter and K_{IK} is a correction factor describing the frequency instability of the calibration item.

The partial derivations of Equ. (6) with respect to the two input variables x_i are the uncertainty contributions $w_S(f_S), w_I(K_{IK})$ which, neglecting covariance, can be summed up according to

$$w(f_X) = \sqrt{\sum_{i=1}^{2} w_i^2}$$

to obtain Total relative measurement uncertainty. Introducing the Coverage factor (k=2), this quantity can be further processed to yield the Expanded measurement uncertainty $W(f_X)$.

2.3 THD (Total harmonic distortion)

This is an engineering definition of THD (Total Harmonic Distortion):

$$THD = \frac{U_{OW}}{U_{GW}} \tag{7}$$

where U_{OW} is the total root-mean-square value of all harmonics and U_{GW} is the rms value of the fundamental wave..

All values are calculated by DFT or FFT from a set of N samples. These samples are provided by the AD converter of the CS18 electronic module on the selected channel.

Neglecting the frequency response of the measurement channel in the frequency range of interest, the absolute expanded measurement uncertainty obtained is w(y) = 0,00025,.

3. Contribution to the measurement uncertainty budget

For a detailed description of the individual contributions and how to measure them, please refer to the details in the comprehensive measurement uncertainty budget [M1]. Table 2 below shows an abstract.

	Defining quantities Correction factors	How to obtain their values
$X_1 = S_S, p_S$	reference standard	from certificate by an accredited calibration laboratory
$X_2 = R$	$R = V_X / V_S$	from previous investigations conducted by SPEKTRA
$X_3 = G$	$G = G_S / G_X$	from previous investigations conducted by SPEKTRA
$X_4 = K_{FPC}$	Free field-pressure conversion	not relevant in this place
$X_5 = K_{IS}$	Instability of BN	from manufacturer's specification
$X_6 = K_{IG}$	Gain coefficients	from previous investigations conducted by SPEKTRA
$X_7 = K_{IR}$	AD converter	from previous investigations conducted by SPEKTRA
$X_8 = K_{TD}$	Temperature deviation	from previous investigations conducted by user
$X_9 = K_{PDM}$	Pressure deviation	from previous investigations conducted by user
$X_{10} = K_{PDC}$	Pressure measurement	from previous investigations conducted by user
$X_{11} = K_{HUM}$	Humidity measurement	from previous investigations conducted by user
$X_{12} = K_{FFC}$	Deviation from free field	not relevant in this place
$X_{13} = K_{POL}$	Polarization voltage	from previous investigations conducted by user
$X_{14} = K_F$	Frequency response	from previous investigations conducted by SPEKTRA
$X_{15} = K_{SUB}$	Substitution	from previous investigations conducted by SPEKTRA
$X_{16} = K_{AD}$	Adaptation	from previous investigations conducted by user
$X_{17} = K_{RES}$	Residual effects	estimated by SPEKTRA, with allowance, if necessary

Table 2: Contributions to the Measurement uncertainty budget and how to obtain their values (see also Table 1)

4. Estimation of total measurement uncertainty for calibration with sinusoidal excitation

4.1 General budget table [1]

Pos.	Quantity X_i	Uncertainty $w(x_i)$	Sensitivity coefficient c_i^*	Uncertainty contribution $w_i(y)$	Variance $w_i^2(y)$
1	S_S	$w(S_S)$	1	$w_{S_S}(S_X)$	$w_{S_S}^2(S_X)$
2	G	$w(G)$	1	$w_G(S_X)$	$w_G^2(S_X)$
3	R	$w(R)$	1	$w_R(S_X)$	$w_R^2(S_X)$
4	K_{FPC}	$w(K_{FPC})$	1	$w_{FPC}(S_X)$	$w_{FPC}^2(S_X)$
5	K_{IS}	$w(K_{IS})$	1	$w_{IS}(S_X)$	$w_{IS}^2(S_X)$
6	K_{IG}	$w(K_{IG})$	1	$w_{IG}(S_X)$	$w_{IG}^2(S_X)$
7	K_{IR}	$w(K_{IR})$	1	$w_{IR}(S_X)$	$w_{IR}^2(S_X)$
8	K_{TD}	$w(K_{TD})$	1	$w_{TD}(S_X)$	$w_{TD}^2(S_X)$
9	K_{PDM}	$w(K_{PDM})$	1	$w_{PDM}(S_X)$	$w_{PDM}^2(S_X)$
10	K_{PDC}	$w(K_{PDC})$	1	$w_{PDC}(S_X)$	$w_{PDC}^2(S_X)$
11	K_{HUM}	$w(K_{HUM})$	1	$w_{HUM}(S_X)$	$w_{HUM}^2(S_X)$
12	K_{FFC}	$w(K_{FFC})$	1	$w_{FFC}(S_X)$	$w_{FFC}^2(S_X)$
13	K_{POL}	$w(K_{POL})$	1	$w_{POL}(S_X)$	$w_{POL}^2(S_X)$
14	K_F	$w(K_F)$	1	$w_F(S_X)$	$w_F^2(S_X)$
15	K_{SUB}	$w(K_{SUB})$	1	$w_{SUB}(S_X)$	$w_{SUB}^2(S_X)$
16	K_{AD}	$w(K_{AD})$	1	$w_{AD}(S_X)$	$w_{AD}^2(S_X)$
17	K_{RES}	$w(K_{RES})$	1	$w_{RES}(S_X)$	$w_{RES}^2(S_X)$
	$Y_X = S_X$, p_X			$w(Y_X) = \sqrt{\sum_{i=1}^{N} w_i^2}$	
	$Y_X = S_X$, p_X			$W(Y_X) = k * w(S_X)$ *mit* $k = 2$	

For the measurement or generation of sound pressure, substitute S_X for p_X

The individual budget tables specific to an individual measurement system are obtained from the general budget table by entering the numerical values allocated to the individual measurement system.

Note: Since all sensitivity coefficients are equal to 1 in all cases, this column is omitted in the individual budget tables specific to a measurement system. Furthermore there is no special column indicating the numerical values of the assumed distribution, which are 2 und $\sqrt{3}$, as there is a separate footnote indicating these values.

[1] in the style of DKD-R 3-1, Part 3 / *) $c_i^* = (x_i / y)\, c_i$

4.2 Special budget table for the LF microphone standard measurement system -

Frequency range 31,5 Hz to 2 kHz (5 kHz**)) / reference standard BN-A-01 / coupler / comparison test

Pos	Defining quantity / influence variable	Uncertainty $w(x_i)$	Distri-bution	Frequency range Hz	Numercial values *) in %
1	Transfer coefficient of standard S_S	$w(S_S)$	Normal	31,5 – 2 k	0,23
2	Pressure-free field conversion	$w(K_{FPC})$	Normal	31,5 – 2 k	not relevant
3	Voltage ratio R	$w(R)$	Normal	31,5 – 2 k	1 (see 3.2.2)
4	Gain ratio G	$w(G)$	Normal	31,5 – 2 k	0,02
5	Long-term stability of S_S	$w(K_{IS})$	Rectang.	31,5 – 2 k	0,033
6	Instability of gain	$w(K_{IG})$	Normal	31,5 – 2 k	0
7	Instability of AD converter	$w(K_{IR})$	Normal	31,5 – 2 k	0
8	Temperature response of S_S	$w(K_{TD})$	Rectang.	31,5 – 2 k	0,02
9	Effect of static pressure on S_S	$w(K_{PDM})$	Rectang.	31,5 – 2 k	0,053
10	Absolute meas. of static pressure	$w(K_{PDC})$	Rectang.	31,5 – 2 k	0
11	Effect of humidity on S_S	$w(K_{HUM})$	Rectang.	31,5 – 2 k	0,002
12	Effect of polarization voltage on S_S	$w(K_{POL})$	Rectang.	31,5 – 2 k	0,06
13	Frequency response of electronic module for substitution K_F	$w(K_F)$	Rectang.	31,5 – 2 k	0,05
14	Influence of substitution K_{SUB}	$w(K_{SUB})$	Rectang.	31,5 – 2 k	not relevant
15	Influence of adaptation K_{AD}	$w(K_{AD})$	Rectang.	31,5 – 2 k	0,6
16	Deviation from free field K_{FFC}	$w(K_{FFC})$	Rectang.	31,5 – 2 k	not relevant
17	Residual effects	$w(K_{RES})$	Rectang.	31,5 – 2 k	0,1

*) Uncertainty contributions $w_i = u_i/y_i$, calculated from estimated limits of input quantities X_i with divisor 2 for normal distribution or divisor √3 for rectangular distribution as assumed distribution models

**) for information only (factory calibration)

Relative ***Total measurement uncertainty w(Y$_X$)*** and ***Expanded measurement uncertainty W(Y$_X$)*** are:

Frequency range Hz	Relative total measurement uncertainty $w(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, dB	Rounded value, dB
31,5 – 2 k	1,7	**3,4**	**0,29**	**0,3**
> 2 k – 5 k	2,9	**5,8**	**0,49**	**0,5*****)

***) estimate, derived from comparison tests in a free sound field

4.3 *Special budget table for the* calibration of microphones and sound level meters by means of an acoustical calibrator

Fixed frequencies 250 Hz and 1.000 Hz / Reference standards BN-A-02 and BN-A-03 / Pressure chamber / Absolute measurement (microphone) or generation (SPM)

Pos.	Defining quantity / influence variable	Uncertainty $w(x_i)$	Distri-bution	Frequency range Hz	Numerical values, % *) microphone | SPM
1a	Sound pressure level p_S of standard 4228 (124 dB)	$w(p_S)$	Normal	250	0,29
1b	Sound pressure level p_S of standard 4231 (94/114 dB)	$w(p_S)$	Normal	1000	0,35
2	Pressure-free field conversion	$w(K_{FPC})$	Normal	250 / 1000	not relevant
3	Voltage measurement R	$w(R)$	Normal	250 / 1000	0,5 | not relevant
4	Gain ratio G	$w(G)$	Normal	250 / 1000	not relevant
5	Long-term stability of p_S	$w(K_{IS})$	Rectang.	250 / 1000	0 (siehe 3.3.2)
6	Instability of gain	$w(K_{IG})$	Normal	250 / 1000	0
7	Instability of AD converter	$w(K_{IR})$	Normal	250 / 1000	0
8	Temperature response of p_S	$w(K_{TD})$	Rectang.	250 / 1000	0,02
9	Effect of static pressure on p_S	$w(K_{PDM})$	Rectang.	250 / 1000	0,053
10	Absolute meas. of static pressure	$w(K_{PDC})$	Rectang.	250 / 1000	0,032
11	Effect of humidity on p_S	$w(K_{HUM})$	Rectang.	250 / 1000	0,002
12	Effect of polarization voltage on p_S	$w(K_{POL})$	Rectang.	250 / 1000	0,06
13	Frequ.resp. of electr. module K_F	$w(K_F)$	Rectang.	250 / 1000	0,05 | not relevant
14	Effect of substitution K_{SUB}	$w(K_{SUB})$	Rectang.	250 / 1000	nicht relevant
15	Effect of adaptation K_{AD}	$w(K_{AD})$	Rectang.	250 / 1000	0,5 | 1,0
16	Deviation from free field K_{FFC}	$w(K_{FFC})$	Rectang.	250 / 1000	not relevant
17	Residual effects	$w(K_{RES})$	Rectang.	250 / 1000	0,1

*) Uncertainty contributions $w_i = u_i/y_i$, calculated from estimated limits of input quantities X_i with divisor 2 for normal distribution or divisor √3 for rectangular distribution as assumed distribution models

Relative ***Total measurement uncertainty w(Y*$_X$*)*** and ***Expanded measurement uncertainty W(Y*$_X$*)*** are:

Type of device	Relative total measurement uncertainty $w(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, dB	Rounded value dB
Microphone	0,94	**1,88**	**0,16**	**0,2**
SPM	1,17	**2,35**	**0,2**	**0,2**

4.4 *Special budget table for the* LF sound calibrator standard measurement system

Fixed frequencies 250 Hz and 1.000 Hz / Reference standard BN-A-01 / Pressure chamber / Substitution measurement

Pos.	Defining quantity / influence variable	Uncertainty $w(x_i)$	Distri-bution	Frequency, Hz	Numerical values, % *)
1a	Sound pressure level p_S of standard BN-A-03 (124 dB)	$w(p_S)$	Normal	250	0,29
1b	Sound pressure level p_S of standard BN-A-02 (94/114 dB)	$w(p_S)$	Normal	1000	0,35
2	Pressure-free field conversion	$w(K_{FPC})$	Normal	250 / 1000	nicht relevant
3	Voltage measurement R	$w(R)$	Normal	250 / 1000	0,03
4	Gain ratio G	$w(G)$	Normal	250 / 1000	0,02
5	Long-term stability of p_S	$w(K_{IS})$	Rectang.	250 / 1000	0 (siehe 3.3.2)
6	Instability of gain	$w(K_{IG})$	Normal	250 / 1000	0
7	Instability of AD converter	$w(K_{IR})$	Normal	250 / 1000	0
8	Temperature response of p_S	$w(K_{TD})$	Rectang.	250 / 1000	0,02
9	Effect of static pressure on p_S	$w(K_{PDM})$	Rectang.	250 / 1000	nicht relevant
10	Absolute meas. of static pressure	$w(K_{PDC})$	Rectang.	250 / 1000	0,05
11	Effect of humidity on p_S	$w(K_{HUM})$	Rectang.	250 / 1000	0,001
12	Effect of polarization voltage on p_S	$w(K_{POL})$	Rectang.	250 / 1000	0,06
13	Frequ.resp. of electr. module for substitution K_F	$w(K_F)$	Rectang.	250 / 1000	0
14	Effect of substitution K_{SUB}	$w(K_{SUB})$	Rectang.	250 / 1000	0 (siehe 3.3.11)
15	Effect of adaptation K_{AD}	$w(K_{AD})$	Rectang.	250 / 1000	0,3
16	Deviation from free field K_{FFC}	$w(K_{FFC})$	Rectang.	250 / 1000	nicht relevant
17	Residual effects	$w(K_{RES})$	Rectang.	250 / 1000	0,1

*) Uncertainty contributions $w_i = u_i/y_i$, calculated from estimated limits of input quantities X_i with divisor 2 for normal distribution or divisor √3 for rectangular distribution as assumed distribution models

Relative ***Total measurement uncertainty $w(Y_X)$*** and ***Expanded measurement uncertainty $W(Y_X)$*** are:

Type of device	Relative total measurement uncertainty $w(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, %	Expanded measure-ment uncertainty $W(Y_X)$, dB	Rounded value dB
BN-A-03	0,77	**1,54**	**0,13**	**0,15**
BN-A-02	0,80	**1,60**	**0,14**	**0,15**

Fixed frequencies 250 Hz and 1.000 Hz / Pressure chamber / Frequency measurement

Pos.	Quantity X_i	Uncertainty $w(x_i)$	Distribution	Uncertainty contribution *) $w_i(y)$
1	Meas. uncertainty of standard	$w(f_S)$	Normal	$0{,}5 \cdot 10^{-8}$
2	Instability of KG	$w(K_{IC})$	Normal	$5 \cdot 10^{-4}$

*) Uncertainty contributions $w_i = u_i/y_i$, calculated from estimated limits of input quantities X_i with divisor 2 for normal distribution or divisor √3 for rectangular distribution as assumed distribution models

Relative ***Total measurement uncertainty*** $w(f_X)$ and ***Expanded measurement uncertainty*** $W(f_X)$ are:

Frequency Hz	Relative total measurement uncertainty $w(f_X)$, %	Expanded measurement uncertainty $W(f_X)$, %	Rounded value %
250	0,05	0,1	0,1
1000	0,05	0,1	0,1

These values also apply to the measurement of multitone calibrators..

Fixed frequencies 250 Hz and 1.000 Hz / Pressure chamber / THD measurement

Pos.	Quantity X_i	Uncertainty $w(x_i)$	Distribution	Uncertainty contribution *) $w_i(y)$
1	Measurement uncertainty for the total of all harmonics	$w(x_1)$	Normal	± 4 LSB = $1{,}2 \cdot 10^{-4}$
2	Measurement uncertainty for the fundamental wave	$w(x_2)$	Normal	± 1 LSB = $3 \cdot 10^{-5}$

*) Uncertainty contributions $w_i = u_i/y_i$, calculated from estimated limits of input quantities X_i with divisor 2 for normal distribution or divisor √3 for rectangular distribution as assumed distribution models

Absolute ***Total measurement uncertainty*** w(THD) and ***Expanded measurement uncertainty*** W(THD) are:

Frequency Hz	Absolute total measurement uncertainty w(THD)	Expanded measurement uncertainty W(THD)	Roughly rounded value
250	0,000125	0,00025	0,001 bzw. 0,1 %
1000	0,000125	0,00025	0,001 bzw. 0,1 %

4.5 Overview of measurement uncertainties of standard measurement systems

Standard measurement system	Frequency range Hz	Expanded meas. uncertainty, dB	Expanded meas. uncertainty of frequ. measurement, %
Microphone / pressure chamber 51AB	31,5 – 2 k > 2 k – 5 k	0,3 0,5	
Sound level meter / pressure chamber 51AB	31,5 – 2 k > 2 k – 5 k	0,3 0,5	
Microphone / pressure chamber	250	0,2	
Microphone / pressure chamber	1000	0,2	
Sound level meter / pressure chamber	250	0,2	
	1000	0,2	
Sound calibrator 124 dB	250	0,15	0,1
Sound calibrator 94 / 114 dB	1.000	0,15	0,1